



# Change One Block

Alter the code and see what happens

Name: \_\_\_\_\_

Date: \_\_\_\_\_

★ I can change one block and tell how the result changes.

## Start with this code

Here is Bit's base code. It starts at 1, then forward 2, then forward 1, so Bit lands on 4. In each exercise you change ONE block, find the new landing number, and tell how the change moved Bit.

## Bit's number path



## Base code (lands on 4)

start at 1

forward 2

forward 1

### 1 Change forward 2 to forward 3

Change the forward 2 block to forward 3. Run the code. What number does Bit land on now? Did Bit land higher or lower than 4?

---

---

### 2 Change the last block to back 1

Go back to the base code. Change the last block (forward 1) to back 1. Run it. What number does Bit land on? Why is it lower?

---

---

### 3 You try — change the start block

Go back to the base code. Change start at 1 to start at 0. PREDICT the new landing number first, then run to check. How did changing the start block move every step?

---

---

---

#### Big idea

Changing even one block changes what the code does. A bigger forward block lands Bit higher; a back block instead of forward lands Bit lower; a different start moves the whole code.



# Change One Block

Quick facilitation notes & answer key

## What this worksheet teaches

Students take one known set of instructions and change ONE block at a time, find the new result, and describe how that single change moved Bit. Change one thing, see what happens. No coding experience needed.

## Before you start (no computer needed)

Paper and pencil only. Optional: a floor number line 0 to 5 so students can run the base code, then re-run after each one-block change.

## How to lead it (about 20 minutes)

1. Show them first (you do). Run the base code to 4. Say: "Now I change ONE block and see what happens."
2. Do one together (we do). Exercise 1: forward 2 → forward 3 makes 5 (one higher — one more step). Exercise 2: last block → back 1 makes 2 (lower — Bit moves the other way).
3. Let them try alone (you do). Exercise 3: start at 1 → start at 0; predict 3, then run; the whole code shifts down by one.
4. Big idea: "Each change moves Bit in a way you can describe — that is reading and changing code."

## Answer key (these are the verified correct answers)

Base code — start 1, forward 2 → 3, forward 1 → 4.

Exercise 1 — forward 2 → forward 3: 1 → 4 → 5. Bit lands on 5, one HIGHER (one extra step).

Exercise 2 — last block → back 1: 1 → 3 → 2. Bit lands on 2, LOWER, because a back block moves Bit the other way.

Exercise 3 — start at 0: 0 → 2 → 3. Bit lands on 3 — one lower at every step because the whole code starts one number earlier.

## If students get stuck — and ways to adjust

Watch for: changing more than one block. Remind students to return to the base code before each exercise — change only one thing.

Make it easier: keep the base code on the floor line and physically swap one card at a time.

Make it harder: ask which single change makes Bit land on 1 (e.g. change forward 2 to back 2).

## Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions someone wrote, change one, and explain what the change did.

## You'll know it worked when

The student finds 5, 2, and 3 and explains each result in terms of the one block that changed.